

ENGR 213 · MASTER COMPANION & EXECUTIVE SUMMARY

Chapter 1: Definitions, Classification, IVPs, & Existence

A comprehensive ground-up companion bridging Dennis G. Zill (*Advanced Engineering Mathematics*, 7th ed., Sections 1.1 & 1.2) with Dr. A. Haghighat's Lecture Notes (Lectures 1 & 2). Every classification rule, interval of definition nuance, IVP condition, and existence & uniqueness proof is fully expanded in plain language with step-by-step worked examples.

Zill 7th Ed. Section 1.1-1.2

Dr. Haghighat Lectures 1 & 2

Concordia University

Complete Step-by-Step Solutions

Picard's Existence & Uniqueness

0 The Big Mental Shift: Rules vs. Stories

TL;DR

In elementary algebra you solve for a static number ($x = 3$). In differential equations, you solve for an entire dynamic function ($y = \phi(x)$). The differential equation is the **rule of change**; solving it means reconstructing the complete **story** that obeys that rule.

A differential equation relates an unknown function $y(x)$ to its rates of change (derivatives $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$, ...). It never gives you coordinates directly; it specifies how the system responds to tiny nudges.

THE CORE ANALOGY · THE PHYSICS ENGINE & THE MOVIE

Think of a differential equation as the **physics engine** of a video game: it dictates how velocity changes with position or how aerodynamic drag counters gravity. It does not decide where the vehicle is right now—it only enforces the laws of movement. Finding a **solution** means hitting "play" and tracing the complete flight path ($y(x)$) that obeys those physical laws.

WHY CONSTANTS (+C) ALWAYS APPEAR

Differentiating wipes out constant vertical offsets: $\frac{d}{dx}(x^2 + 5) = 2x$ and $\frac{d}{dx}(x^2 - 100) = 2x$. Because differentiating destroys constant information, integrating backwards can never recover what specific constant was originally there. It leaves an arbitrary constant C , handing back an **infinite family of parallel curves**.

DEPENDENT VS. INDEPENDENT VARIABLES

Inspect the derivative symbol $\frac{dy}{dx}$ (or $\frac{du}{dt}$):

- **Numerator (y):** The *dependent* variable (the quantity being tracked).
- **Denominator (x or t):** The *independent* variable (the coordinate or time clock driving the system).

Physical Engineering Systems Modeled by DEs (Dr. Haghighat Lecture 1, Slide 4 & Zill Section 1.3)

Physical System	Governing DE	Plain-English Physical Meaning
Newton's Law of Cooling	$\frac{dT}{dt} = -k(T - T_m)$	Cooling rate ($\frac{dT}{dt}$) is proportional to the difference between current temperature T and ambient room temperature T_m .
Newton's 2nd Law (Motion with Drag)	$m\frac{dv}{dt} = mg - kv$	Net force equals mass times acceleration ($\frac{dv}{dt}$). Downward gravity mg is opposed by air resistance proportional to velocity kv .
Logistic Population Dynamics	$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$	Growth is exponential initially (rP), but tapers smoothly to zero as population P approaches environmental carrying capacity K .
First-Order Chemical Kinetics	$\frac{dA}{dt} = -kA$	The reaction rate of substance A is proportional to the concentration of reactant molecules remaining to collide.

THE 3-STEP CANDIDATE SOLUTION VERIFICATION PROTOCOL

To verify that $y = \phi(x)$ solves an n -th order ODE on interval I :

1. **Compute Derivatives:** Differentiate $\phi(x)$ systematically up to order n ($\phi', \phi'', \dots, \phi^{(n)}$).
2. **Substitute Both Sides:** Insert $\phi(x)$ and all its derivatives into the Left-Hand Side (LHS) and Right-Hand Side (RHS) of the ODE.
3. **Confirm Algebraic Identity:** Show that $\text{LHS} \equiv \text{RHS}$ for every single $x \in I$. If it holds identically, it is a valid solution.

1

Classification of Differential Equations: The 3 Golden Axes

TL;DR

Every differential equation is classified by **Type** (ODE vs. PDE), **Order** (highest derivative present), and **Linearity** (degree 1 in y and its derivatives, coefficients depending only on x). Classification determines which analytical tools will work.

AXIS 1: CLASSIFICATION BY TYPE

- **ODE (Ordinary):** Derivatives involve a *single* independent variable (e.g., $y'(x) + 5y = e^x$).
- **PDE (Partial):** Derivatives involve *two or more* independent variables (e.g., $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$).

AXIS 2: ORDER & NORMAL FORM

- **Order:** The order of the *highest derivative* present in the equation.
- **Normal Form:** Solved for highest derivative: $\frac{d^ny}{dx^n} = f(x, y, y', \dots, y^{(n-1)})$.

EXAM TRAP CRITICAL EXAM TRAP: ORDER VS. DEGREE CONFUSION

Look at: $\left(\frac{dy}{dx}\right)^5 + 4y = \sin(x)$.
Students often rush and call this "5th order." **Wrong!** The derivative is only $\frac{dy}{dx}$ (order 1). The exponent 5 is the *degree*. This is a **first-order, non-linear ODE!**

AXIS 3: CLASSIFICATION BY LINEARITY (THE 3 MANDATORY RULES · ZILL SECTION 1.1, P. 53)

An n -th order ODE is **linear** if and only if it can be written strictly in the form:

$$a_n(x)\frac{d^ny}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \dots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

An ODE is linear if and only if it passes **all three** strict tests:

- Rule 1 (First-Degree):** The dependent variable y and all its derivatives $y', y'', \dots$ appear strictly to the first power (no y^2 , $\sqrt{y}$, $(y')^3$).
- Rule 2 (Pure- x Coefficients):** The coefficients $a_i(x)$ depend *strictly* on the independent variable x (or constants). No terms like $y\frac{dy}{dx}$!
- Rule 3 (No Nonlinear Functions of y):** No $\sin(y)$, e^y , $\ln(y)$, $\cos(y')$.

KEY RULE THE "WILD X" FREEDOM RULE

Linearity places strict handcuffs on the dependent variable y . But the independent variable x has complete freedom! An equation like $x^4 \cos(x)y'' + \ln(x)y' + e^xy = \arctan(x)$ is **100% linear** because every appearance of y, y', y'' obeys the three rules!

Master Classification Drill: Testing All Combinations

Differential Equation	Type	Order	Linearity	Exact Architectural Reason
$4x\frac{dy}{dx} + y = x^3$	ODE	1st	✓ Linear	y and y' have exponent 1; coefficients $(4x, 1)$ depend only on x .
$y'' - 2y' + y = 0$	ODE	2nd	✓ Linear	Constant coefficients $(1, -2, 1)$; all derivatives have power 1.
$\frac{d^3y}{dx^3} + x\frac{dy}{dx} + (\cos x)y = e^x$	ODE	3rd	✓ Linear	Highest derivative is y''' ; nonlinear trig function is on x , which is allowed!
$(y')^2 + y = x$	ODE	1st	✗ Nonlinear	First derivative is squared: $(y')^2$. Violates Rule 1.
$y\frac{dy}{dx} + 2y = 1$	ODE	1st	✗ Nonlinear	Coefficient of y' contains y (yy'). Violates Rule 2.
$\frac{d^2y}{dx^2} + \sin(y) = 0$	ODE	2nd	✗ Nonlinear	Nonlinear sine function applied directly to dependent variable y . Violates Rule 3.
$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial t^2} = 0$	PDE	2nd	✓ Linear	Partial derivatives with respect to two independent variables (x and t).

2 Solutions, Domains of Validity, & Solution Types

TL;DR

A solution is a function $y = \phi(x)$ that satisfies the ODE identically on a specific connected **interval of definition** (I). A formula without its valid domain is incomplete. Solutions can be explicit or implicit, form n -parameter families, or exist as mysterious "singular solutions" that elude the family formula.

FORMAL DEFINITION · SOLUTION & INTERVAL OF DEFINITION (ZILL SECTION 1.1, P. 57)

A function ϕ defined on an interval I and possessing at least n continuous derivatives on I is a **solution** of an n -th order ODE on I if, when $y = \phi(x)$ and its derivatives are substituted into the equation, the equation reduces to an identity for all $x \in I$.

The interval I is called the **interval of definition**, **domain of validity**, or **interval of existence**.

EXAM TRAP THE DISCONTINUOUS ASYMPTOTE TRAP: INTERVALS CANNOT JUMP

An interval I in calculus is strictly a **connected** set: (a, b) , $[a, b]$, $(-\infty, \infty)$, etc. It can **never** be a disconnected union like $(-\infty, 0) \cup (0, \infty)$.

Concrete Example: Consider $xy' + y = 0$. The function $y = \frac{1}{x}$ satisfies this equation since $x(-\frac{1}{x^2}) + \frac{1}{x} = 0$.

However, a solution curve cannot leap across the vertical asymptote at $x = 0$. If an initial condition is given at $x_0 = 2$, the interval of definition is strictly $I = (0, \infty)$. If given at $x_0 = -3$, $I = (-\infty, 0)$. It can *never* include $x = 0$.

EXPLICIT VS. IMPLICIT SOLUTIONS

• **Explicit:** y is isolated: $y = \phi(x)$ (e.g., $y = e^{3x} + \cos x$).

• **Implicit:** A relation $G(x, y) = 0$ defining y implicitly.

The Circle Example: $x^2 + y^2 = 25$ implicitly solves $\frac{dy}{dx} = -\frac{x}{y}$. It defines two explicit branches: $y = +\sqrt{25 - x^2}$ and $y = -\sqrt{25 - x^2}$, both valid on open interval $(-5, 5)$. (At ± 5 , slopes are vertical, so endpoints are excluded).

FAMILIES VS. SINGULAR SOLUTIONS

• **n -Parameter Family:** An n -th order ODE produces n arbitrary constants: $y = \Phi(x, c_1, \dots, c_n)$.

• **Trivial Solution:** Constant zero solution $y(x) \equiv 0$ satisfying a homogeneous DE.

• **Singular Solution (Zill's Insight):** A valid solution that *cannot* be obtained by choosing any value of c_i in the family formula (envelope curves).

DEEP DIVE Zill's Classic Singular Solution: The Envelope Parabola (Zill Section 1.1, p. 61)

Consider the differential equation: $\left(\frac{dy}{dx}\right)^2 - 4y = 0 \implies \frac{dy}{dx} = 2\sqrt{y}$.

FAMILY Separating variables: $\frac{dy}{2\sqrt{y}} = dx \implies \sqrt{y} = x + c \implies y = (x + c)^2$. 1-parameter family of shifting parabolas.

SINGULAR Notice that the flat horizontal line $y(x) \equiv 0$ also solves the DE identically: $\left(\frac{d}{dx}(0)\right)^2 - 4(0) = 0 - 0 = 0$.

THE CATCH Can you choose any real number for c in the family $y = (x + c)^2$ to make it equal zero everywhere for all x ? **No!** The parabola $(x + c)^2$ only touches zero at $x = -c$. Therefore, $y \equiv 0$ is a **singular solution**—the envelope tangent to every parabola in the family.

WORKED EXAMPLE 1 Step-by-Step Verification of a 2nd-Order Solution (Dr. Haghighat Lecture 1, Slide 19)

Problem: Verify that $y = c_1 e^x + c_2 x e^x$ is a 2-parameter family of solutions to $y'' - 2y' + y = 0$ on $(-\infty, \infty)$.

STEP I $y' = \frac{d}{dx}(c_1 e^x) + \frac{d}{dx}(c_2 x e^x) = c_1 e^x + c_2(e^x + x e^x) = (c_1 + c_2)e^x + c_2 x e^x$.

STEP II $y'' = \frac{d}{dx}[(c_1 + c_2)e^x + c_2 x e^x] = (c_1 + c_2)e^x + c_2(e^x + x e^x) = (c_1 + 2c_2)e^x + c_2 x e^x$.

STEP III LHS = $y'' - 2y' + y = [(c_1 + 2c_2) - 2(c_1 + c_2) + c_1]e^x + [c_2 - 2c_2 + c_2]x e^x = 0 \cdot e^x + 0 \cdot x e^x \equiv 0 = \text{RHS}$.

VERIFIED Identity holds for all $x \in (-\infty, \infty)$ and all arbitrary constants c_1, c_2 .

WORKED EXAMPLE 2 Finding the Maximal Interval of Definition (I): Nonlinear Singularities (Zill Section 1.1 Ex 5)

Problem: Solve $\frac{dy}{dx} = y^2$ with initial condition $y(0) = 1$, and determine its maximal interval of definition I .

SOLVE Separating variables: $\int y^{-2} dy = \int dx \implies -\frac{1}{y} = x + C \implies y(x) = -\frac{1}{x+C}$.

APPLY IC Apply $y(0) = 1$: $1 = -\frac{1}{0+C} \implies C = -1 \implies y(x) = \frac{1}{1-x}$.

INTERVAL The function has a vertical asymptote at $x = 1$. The domain of the formula is $(-\infty, 1) \cup (1, \infty)$. But the solution to an IVP must be defined on a **single connected interval containing** $x_0 = 0$. Because $0 \in (-\infty, 1)$, the interval of definition is strictly $I = (-\infty, 1)$.

KEY TAKEAWAY Even though the DE $\frac{dy}{dx} = y^2$ is perfectly continuous everywhere, nonlinear solutions can explode to infinity in finite time. The interval of definition depends directly on the initial condition!

3 Initial-Value Problems (IVPs): Pinpointing One Curve

TL;DR

An **Initial-Value Problem (IVP)** pairs an n -th order DE with n auxiliary conditions at **one single initial point** x_0 ($y(x_0) = y_0, y'(x_0) = y_1, \dots$). This single snapshot collapses the infinite family of curves into one specific trajectory.

INITIAL-VALUE PROBLEM (IVP)

All conditions are specified at the **exact same point** x_0 :

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad \dots, \quad y^{(n-1)}(x_0) = y_{n-1}$$

Models evolutionary time-dependent problems starting from $t = t_0$.

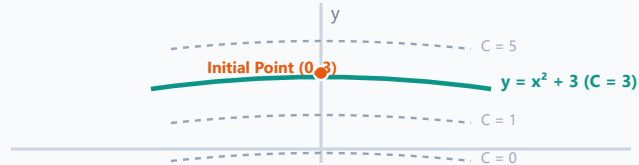
BOUNDARY-VALUE PROBLEM (BVP)

Conditions are specified at **different boundary points** $x_a \neq x_b$:

$$y(0) = 0, \quad y(L) = 0$$

Models static spatial equilibria (e.g., beam deflection between supports).

VISUALIZING AN IVP: INFINITE FAMILY $Y = X^2 + C$ COLLAPSING TO $C = 3$ AT $(0, 3)$



1ST-ORDER IVP BASELINE CASE

Solve $\frac{dy}{dx} = 2x, y(0) = 3$: General Solution: $y(x) = \int 2x dx = x^2 + C$. Apply $y(0) = 3 \implies 3 = (0)^2 + C \implies C = 3$.

Unique Particular Solution: $y(x) = x^2 + 3$, valid for all $x \in (-\infty, \infty)$.

WORKED EXAMPLE Solving a 2nd-Order IVP (Zill Section 1.2 Ex 3, p. 91 & Dr. Haghighat Lecture 2, Slide 5)

Problem: Given $x(t) = c_1 \cos(4t) + c_2 \sin(4t)$ is the general solution to $x'' + 16x = 0$, solve the initial-value problem:

$$x'' + 16x = 0, \quad x\left(\frac{\pi}{2}\right) = -2, \quad x'\left(\frac{\pi}{2}\right) = 1$$

STEP I Differentiate using Chain Rule: $x'(t) = \frac{d}{dt}[c_1 \cos(4t) + c_2 \sin(4t)] = -4c_1 \sin(4t) + 4c_2 \cos(4t)$.

STEP II Apply $x(\frac{\pi}{2}) = -2$: $x(\frac{\pi}{2}) = c_1 \cos(2\pi) + c_2 \sin(2\pi) = c_1(1) + c_2(0) = -2 \implies c_1 = -2$.

STEP III Apply $x'(\frac{\pi}{2}) = 1$: $x'(\frac{\pi}{2}) = -4c_1 \sin(2\pi) + 4c_2 \cos(2\pi) = 4c_2(1) = 1 \implies c_2 = \frac{1}{4}$.

PARTICULAR SOLUTION $x(t) = -2 \cos(4t) + \frac{1}{4} \sin(4t)$ valid on $(-\infty, \infty)$

ENGINEERING IVP: FALLING BODY WITH AIR DRAG (ZILL SECTION 1.3)

$$m \frac{dv}{dt} = mg - kv, \quad v(0) = 0.$$

1. Linear standard form: $\frac{dv}{dt} + \frac{k}{m}v = g$.

2. General solution: $v(t) = \frac{mg}{k} + Ce^{-\frac{k}{m}t}$.

3. Apply $v(0) = 0 \implies C = -\frac{mg}{k}$.

Particular Solution: $v(t) = \frac{mg}{k} \left(1 - e^{-\frac{k}{m}t}\right)$.

Terminal Velocity: As $t \rightarrow \infty, v \rightarrow v_\infty = \frac{mg}{k}$.

BVP BEHAVIOR: 0, 1, OR INFINITE SOLUTIONS!

Consider $y'' + y = 0 \implies y = c_1 \cos x + c_2 \sin x$.

• **Unique IVP:** $y(0) = 0, y'(0) = 1 \implies y = \sin x$ (Always 1 unique solution!).

• **BVP with 0 Solutions:** $y(0) = 0, y(\pi) = 1 \implies c_1 = 0$, but $y(\pi) = 0 \neq 1$. Impossible!

• **BVP with ∞ Solutions:** $y(0) = 0, y(\pi) = 0 \implies c_1 = 0$, and $y(\pi) = c_2(0) = 0$ holds for *any* $c_2 \in \mathbb{R}$!

4 Existence & Uniqueness Theorem: The Two-Stage Inspection

TL;DR

Before investing hours trying to solve an IVP $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$, Theorem 1.2.1 provides a two-stage safety test:

1. Continuity of $f(x, y)$ near (x_0, y_0) guarantees that at least one solution **exists**.
2. Continuity of the partial derivative $\frac{\partial f}{\partial y}$ guarantees that the solution is **unique**.

THEOREM 1.2.1 · PICARD'S EXISTENCE AND UNIQUENESS THEOREM (ZILL SECTION 1.2, P. 94)

Let R be a rectangular region in the xy -plane: $a \leq x \leq b$, $c \leq y \leq d$ containing the initial point (x_0, y_0) in its interior:

- **Condition 1 (Existence):** If $f(x, y)$ is continuous on R , then there exists some interval $I_0 = (x_0 - h, x_0 + h)$ with $h > 0$ contained in $[a, b]$ and a function $y(x)$ defined on I_0 that solves the IVP.
- **Condition 2 (Uniqueness):** If, in addition, $\frac{\partial f}{\partial y}$ is also continuous on R , then the solution $y(x)$ is **unique** on I_0 .

THE ANALOGY · PAVED TRAILS VS. DIVERGING PATHS

Picture standing at a trailhead (x_0, y_0) . Condition 1 checks whether the ground exists under your feet (is f continuous?). If yes, you can walk forward—a path exists. Condition 2 checks whether the trail has clear guardrails (is $\frac{\partial f}{\partial y}$ continuous?). If yes, there is only one legal path to walk. But if $\frac{\partial f}{\partial y}$ spikes to infinity, the trail fractures into multiple legal paths starting from the exact same spot!

THEOREM 1.2.1 TWO-STAGE INSPECTION PROTOCOL



DIRECTION FIELDS (SLOPE FIELDS) & ISOCINES (DR. HAGHIGHAT LECTURE 2, SLIDES 6–7)

- **Direction Field:** A visual grid of tiny segments with slope $\frac{dy}{dx} = f(x, y)$ at every point (x, y) . Solution curves follow these segments like leaves drifting in a river current.
- **Method of Isoclines:** Curves along which the slope is constant: $f(x, y) = c$. Sketching several isoclines reveals the entire flow field without solving analytically.
- **Geometric Test for Uniqueness:** If f and $\frac{\partial f}{\partial y}$ are continuous, direction field slopes change smoothly, so streamlines never intersect. Intersection implies a breakdown of uniqueness!

WORKED EXAMPLE Testing Regions of Existence & Uniqueness (Zill Section 1.2 Ex 2, p. 92)

Problem: Determine the region of the xy -plane where the IVP $\frac{dy}{dx} = \frac{x^2}{1-y^2}$, $y(0) = 0$ is guaranteed a unique solution.

1. **Examine $f(x, y) = \frac{x^2}{1-y^2}$ (Existence):** Continuous everywhere except where denominator vanishes: $1 - y^2 = 0 \implies y = \pm 1$.
 2. **Examine $\frac{\partial f}{\partial y}$ (Uniqueness):** $\frac{\partial f}{\partial y} = \frac{x^2(2y)}{(1-y^2)^2}$, which also has discontinuities strictly along lines $y = \pm 1$.
 3. **Apply Initial Condition $(0, 0)$:** Since $y_0 = 0 \in (-1, 1)$, we can draw an open rectangle R around $(0, 0)$ completely inside $-1 < y < 1$.
- Conclusion:** Theorem 1.2.1 guarantees a **unique solution** on some open interval I_0 containing $x_0 = 0$.

Quick-Lookup: Where Uniqueness Fails for Classic Exam Functions

Function $f(x, y)$	Partial Derivative $\frac{\partial f}{\partial y}$	Where Theorem 1.2.1 Guarantees vs. Fails
$f(x, y) = xy^{1/2}$	$\frac{\partial f}{\partial y} = \frac{x}{2\sqrt{y}}$	Guaranteed unique for $y > 0$. Uniqueness fails along $y = 0$ (x -axis).
$f(x, y) = y^{2/3}$	$\frac{\partial f}{\partial y} = \frac{2}{3y^{1/3}}$	f continuous everywhere on $\mathbb{R}^2$, but uniqueness fails along $y = 0$.
$f(x, y) = \frac{y}{x-3}$	$\frac{\partial f}{\partial y} = \frac{1}{x-3}$	Both existence and uniqueness fail along vertical line $x = 3$.
$f(x, y) = \sqrt{4 - x^2 - y^2}$	$\frac{\partial f}{\partial y} = \frac{-y}{\sqrt{4 - x^2 - y^2}}$	Unique strictly inside disk $x^2 + y^2 < 4$. Fails on boundary circle $x^2 + y^2 = 4$.

CLASSIC EXAM CASE When Uniqueness Fails: The Bifurcating Solution (Zill Section 1.2 Ex 4, p. 93 & Dr. Haghighat Slide 8)**Problem: Apply Theorem 1.2.1 to test existence and uniqueness for the IVP:**

$$\frac{dy}{dx} = xy^{1/2}, \quad y(0) = 0$$

STEP I Stage 1 (Existence): $f(x, y) = x\sqrt{y}$ is continuous for $y \geq 0$, including at $(0, 0)$. $\implies$ A solution exists! The trivial function $\mathbf{y}(\mathbf{x}) \equiv \mathbf{0}$ works: $0' = x\sqrt{0} \implies 0 \equiv 0$ and $y(0) = 0$.**STEP II Stage 2 (Uniqueness):** Compute partial derivative $\frac{\partial f}{\partial y}$:

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(xy^{1/2}) = \frac{x}{2\sqrt{y}}$$

As $y \rightarrow 0^+$, $\frac{\partial f}{\partial y}$ divides by zero and **blows up to infinity!** It is undefined along $y = 0$.Any open rectangle enclosing $(0, 0)$ contains points where $\frac{\partial f}{\partial y}$ is discontinuous. **Uniqueness is NOT guaranteed!****STEP III Derive Second Solution (Separation of Variables):** For $y > 0$, $\frac{dy}{\sqrt{y}} = x dx \implies \int y^{-1/2} dy = \int x dx \implies 2\sqrt{y} = \frac{1}{2}x^2 + c_1 \implies y = (\frac{1}{4}x^2 + c)^2$.Apply $y(0) = 0 \implies c = 0 \implies \mathbf{y}(\mathbf{x}) = \frac{1}{16}\mathbf{x}^4$.**STEP IV Verification of Second Solution:** $y(0) = \frac{1}{16}(0)^4 = 0$.

$$y' = \frac{4}{16}x^3 = \frac{1}{4}x^3, \text{ and } x\sqrt{y} = x\sqrt{\frac{1}{16}x^4} = \frac{1}{4}x^3 \implies y' \equiv x\sqrt{y}.$$

RESULT Through $(0, 0)$, there are **two distinct solutions**: $y_1(x) = 0$ and $y_2(x) = \frac{1}{16}x^4$. Uniqueness failed completely.**GEOMETRY OF NON-UNIQUENESS: TWO VALID TRAJECTORIES BRANCHING AT $(0, 0)$** **ZILL'S SHOCKING REMARK: INFINITELY MANY PIECEWISE SOLUTIONS!**Because $y = 0$ and $y = \frac{1}{16}x^4$ touch tangentially with continuous derivative at $x = 0$, we can glue them together! For any constant $c \geq 0$:

$$y(x) = \begin{cases} 0, & x < c \\ \frac{1}{16}(x^2 - c^2)^2, & x \geq c \end{cases}$$

is continuously differentiable and satisfies $y' = x\sqrt{y}$, $y(0) = 0$ everywhere! Uniqueness doesn't just fail with two solutions—it yields **uncountably infinitely many valid solutions!****CONTRAST: INITIAL POINT AT $(0, 4)$** At $(0, 4)$, $y_0 = 4 > 0$. We can draw a rectangle R entirely above the x -axis ($y \in [2, 6]$). There, f and $\frac{\partial f}{\partial y} = \frac{x}{2\sqrt{y}}$ are **both continuous!** Uniqueness is guaranteed: $y(x) = (\frac{1}{4}x^2 + 2)^2$.**ANOTHER CLASSIC CASE: $DY/DX = 3Y^{2/3}$** For $\frac{dy}{dx} = 3y^{2/3}$, $y(0) = 0$: $\frac{\partial f}{\partial y} = \frac{2}{y^{1/3}}$ divides by zero at $y = 0$. Both $y(x) \equiv 0$ and $y(x) = x^3$ are valid solutions satisfying $y(0) = 0$!**THE 3 GOLDEN RULES OF PICARD'S THEOREM FOR MIDTERM 1**

- Sufficient, Not Necessary:** Theorem 1.2.1 is a one-way test. If conditions hold $\implies$ existence/uniqueness guaranteed. If conditions fail, a unique solution might still exist by coincidence, but Theorem 1.2.1 cannot be used to prove it.
- Streamlines Cannot Cross:** In any region where both f and $\frac{\partial f}{\partial y}$ are continuous, two solution curves can **never intersect or branch**. Intersection immediately signals a failure of uniqueness.
- Test the Trivial Solution First:** When asked about non-uniqueness at $(x_0, 0)$, always check if $y(x) \equiv 0$ works. It almost always serves as Solution 1!

5 Master Cross-Reference Matrix & Exam Pitfall Guide

TL;DR

This matrix correlates Dennis G. Zill's standard textbook sections with Dr. Haghighat's lecture slides, followed by the five most dangerous traps that cost Concordia engineering students marks on Midterm 1.

Core Concept	Zill 7th Ed. Reference	Dr. Haghighat Slides	Key Exam Takeaway / Strategic Rule
Classification (Type, Order, Linearity)	Section 1.1, pp. 49–56	Lecture 1, Slides 6–11	Linearity is strictly about y and its derivatives. Wild nonlinear coefficients in x (e.g. $x^4 \sin x$) remain 100% linear!
Physical Systems Modeling	Section 1.3, pp. 68–80	Lecture 1, Slide 4	Newton's cooling ($\frac{dT}{dt} = -k(T - T_m)$) and drag ($m\frac{dv}{dt} = mg - kv$) are classic exam modeling setups.
Interval of Definition (I)	Section 1.1, pp. 57–60	Lecture 1, Slide 15	A solution curve cannot cross vertical asymptotes. Interval I must be connected and contain initial point x_0 .
Explicit vs. Implicit Solutions	Section 1.1, pp. 62–64	Lecture 1, Slide 16	Circles $x^2 + y^2 = C$ are implicit. Choosing an explicit branch requires matching the sign of initial coordinate y_0 .
Singular Solutions	Section 1.1, pp. 60–61	Lecture 1, Slide 18	Valid solutions that cannot be generated by choosing any value of C in the general family formula (envelope curves).
2nd-Order IVP	Section 1.2, Ex 3, p. 91	Lecture 2, Slide 5	Both conditions must be evaluated at the <i>exact same point</i> x_0 to form an IVP; otherwise, it is a BVP.
Existence & Uniqueness	Section 1.2, Thm 1.2.1, p. 94	Lecture 2, Slides 7–8	Always check $\frac{\partial f}{\partial y}$. If $\frac{\partial f}{\partial y}$ divides by zero at the initial condition, uniqueness fails!

5 Deadly Exam Pitfalls to Evade on Midterm 1

TRAP 1 THE ORDER VS. DEGREE CONFUSION

$(y')^4 + y = 0$ has power 4, but it is a **first-order** equation because the highest derivative is only y' . Never confuse the exponent on a derivative with the order of differentiation!

TRAP 2 THE NONLINEARITY DOUBLE STANDARD

Students see $\sin(x)y' + y = 0$ and call it nonlinear because of $\sin(x)$. **It is completely linear!** Nonlinearity is only triggered if y or its derivatives are inside the nonlinear function: $\sin(y) \implies$ nonlinear, but $\sin(x)y \implies$ linear.

TRAP 3 FORGETTING THAT INTERVALS CANNOT JUMP HOLES

If asked for the interval of definition for $y = \frac{1}{x-3}$ with $y(1) = -0.5$, the answer is strictly $(-\infty, 3)$. Writing $(-\infty, 3) \cup (3, \infty)$ loses marks because solution curves must be differentiable on a single connected interval.

TRAP 4 CONFUSING IVP WITH BVP CONDITIONS

For a second-order equation, conditions given at two different points ($y(0) = 1$ and $y(2) = 5$) form a **Boundary-Value Problem (BVP)**. Theorem 1.2.1 applies *strictly* to IVPs where all conditions reside at the identical point x_0 .

TRAP 5 ASSUMING EXISTENCE GUARANTEES UNIQUENESS

Continuity of $f(x, y)$ only buys you **existence** (at least one solution). You must independently evaluate $\frac{\partial f}{\partial y}$ at (x_0, y_0) to prove **uniqueness**. Never stop after checking $f(x, y)$!

6

Executive Summary & Comprehensive Review Checklist

TL;DR

Use this final checklist and formula synthesis before sitting for any exam or handing in assignments covering Chapter 1 of ENGR 213.

The 5-Point Pre-Submission Exam Checklist:

✓ **Classification:** Check linearity strictly on y and its derivatives. Coefficients in x can be nonlinear without affecting linearity.

✓ **Order:** Isolate the highest derivative without confusing degree/powers (e.g., $(y')^3$ is 1st order).

✓ **Interval:** State the largest connected open interval I containing x_0 where the solution and its derivatives remain continuous.

✓ **Initial Conditions:** Confirm that all n conditions for an n -th order IVP are evaluated at the exact same x_0 .

✓ **Existence & Uniqueness:** Verify both $f(x, y)$ and $\frac{\partial f}{\partial y}$ for continuity across an open rectangle enclosing (x_0, y_0) .

Core Mathematical Formulation Quick-Reference

GENERAL LINEAR N -TH ORDER ODE

$$a_n(x)y^{(n)} + \cdots + a_1(x)y' + a_0(x)y = g(x)$$

Standard Form (divide by $a_n(x)$):

$$y^{(n)} + P_{n-1}(x)y^{(n-1)} + \cdots + P_0(x)y = f(x)$$

PICARD'S THEOREM (THEOREM 1.2.1)

For $\frac{dy}{dx} = f(x, y), y(x_0) = y_0$:

• f continuous on $R \implies$ **Existence.**

• $\frac{\partial f}{\partial y}$ continuous on $R \implies$ **Uniqueness.**

N -TH ORDER INITIAL-VALUE PROBLEM

Solve: $\frac{d^N y}{dx^N} = f(x, y, y', \dots, y^{(n-1)})$

Subject to:

$$y(x_0) = y_0, y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1}$$

LOOKING AHEAD: CHAPTER 2 METHODS

• **Separable (Section 2.2):** $g(y)dy = f(x)dx$.

• **Linear 1st-Order (Section 2.3):** $\mu(x) = e^{\int P(x)dx}$.

• **Exact Equations (Section 2.4):** $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Essential Calculus Checklist for Verification & IVP Steps

Operation	Calculus Formula	Common Pitfall to Avoid
Product Rule	$\frac{d}{dx}[u(x)v(x)] = u'v + uv'$	Forgetting to differentiate both factors in $xe^x \implies e^x + xe^x$.
Chain Rule	$\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$	Forgetting inner derivative in $\cos(4t) \implies -4\sin(4t)$.
Log Integral	$\int \frac{1}{x}dx = \ln x + C$	Dropping absolute value signs, altering interval of definition.

MIDTERM 1 SCORE-MAXIMIZATION PROTOCOL

1. **Classify Instantly (30s):** Verify linearity solely on y and its derivatives. 2. **Identify Interval Before Algebra:** Check denominators and square roots for domain limits. 3. **Carry Constants Through Steps:** Never tack $+C$ on at the very end. 4. **LHS = RHS Verification:** Spend 60 seconds plugging solutions back into the original ODE for guaranteed full marks.

✓ **TYING IT ALL TOGETHER**

Every differential equation is a puzzle connecting rates of change. Classification tells you which box of tools to open; the interval of definition sets the boundaries of reality; the initial conditions pick the exact trajectory from the family; and Picard's theorem ensures the trajectory is solid, unambiguous, and dependable.

ENGR 213: Ordinary Differential Equations · Comprehensive Study Companion · Concordia University

Based on Dennis G. Zill (*Advanced Engineering Mathematics*, 7th ed., Chapter 1) & Dr. A. Haghighat M. Lecture Notes (Lectures 1 & 2).

Authored with complete step-by-step expansions, pedagogical analogies, and original diagrams.

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